2 . DDL Commands with Constraints - PRIMARY, FOREIGN KEY, UNIQUE, CHECK, DEFAULT, NOT NULL

```
mysql> create table dept ( dept_id int primary key , dept_name varchar(50) unique ) ;
Query OK, 0 rows affected (2.31 sec sec)
mysql> create table learners ( l_id int primary key , l_name varchar(30) not null , l_age int
  check ( l_age >= 18 ) , l_city varchar(20) default 'chennai' , email varchar(40) , dept_id
  int ) ;
Query OK, 0 rows affected (3.05 sec sec)
mysql> alter table learners add constraint fk_dept foreign key ( dept_id ) references dept (
  dept_id ) ;
Query OK, 0 rows affected (3.42 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> insert into learners ( l_id , l_name , l_age , email , dept_id ) values ( 1 , "arun" ,
  16 , "arun07@gmail.com" , 201 ) ;
ERROR 3819 (HY000): Check constraint 'learners_chk_1' is violated.
mysql> insert into learners ( l_id , l_name , l_age , email , dept_id ) values ( 1 , "arun" ,
  19 , "arun07@gmail.com" , 201 ) ;
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails
  (`college`.`learners`, CONSTRAINT `fk_dept` FOREIGN KEY (`dept_id`) REFERENCES `dept`
  (`dept_id`))
mysql> insert into dept values ( 201 , "mechanical" ) , ( 202 , "eee" ) , ( 203 , "it" ) ;
Query OK, 3 rows affected (1.08 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> insert into learners ( l_id , l_name , l_age , email , dept_id ) values ( 1 , "arun" ,
  19 , "arun07@gmail.com" , 201 ) ;
Query OK, 1 row affected (0.29 sec)

mysql> insert into learners ( l_id , l_name , l_age , email , dept_id ) values ( 2 , "meena" ,
  20 , "meena@gmail.com" , 202 ) ;
Query OK, 1 row affected (0.18 sec)
```

```
mysql> insert into learners ( l_id , l_age , email , dept_id ) values ( 3 , 21 , "x@gmail.com"
  , 203 ) ;
ERROR 1364 (HY000): Field 'l_name' doesn't have a default value
mysql> insert into dept values ( 204 , "it" ) ;
ERROR 1062 (23000): Duplicate entry 'it' for key 'dept.dept_name'
mysql> select * from dept ;
+---------+------------+
| dept_id | dept_name  |
+---------+------------+
|     201 | mechanical |
|     202 | eee        |
|     203 | it         |
+---------+------------+
3 rows in set (0.00 sec)

mysql> select * from learners ;
+------+--------+-------+---------+------------------+---------+
| l_id | l_name | l_age | l_city  | email            | dept_id |
+------+--------+-------+---------+------------------+---------+
|    1 | arun   |    19 | chennai | arun07@gmail.com |     201 |
|    2 | meena  |    20 | chennai | meena@gmail.com  |     202 |
+------+--------+-------+---------+------------------+---------+
2 rows in set (0.00 sec)
```